
The Maieutic Reversal Protocol

Extracting Latent Ontological Priors from LLM Judges

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Abstract

The "LLM-as-a-judge" paradigm suffers from opacity regarding the evaluator's own biases. We introduce the Maieutic Reversal Protocol, a formalization of Socratic questioning designed to extract the latent ontological priors of LLM judges. By recursively questioning the evaluator, we expose hidden assumptions regarding Human-System (HS) versus Human-Resource (HR) orientations.

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1 Introduction

[cite_start]Motivation : Ensuring transparency in automated evaluation pipelines[cite : 1, 4].

2 Main Contributions

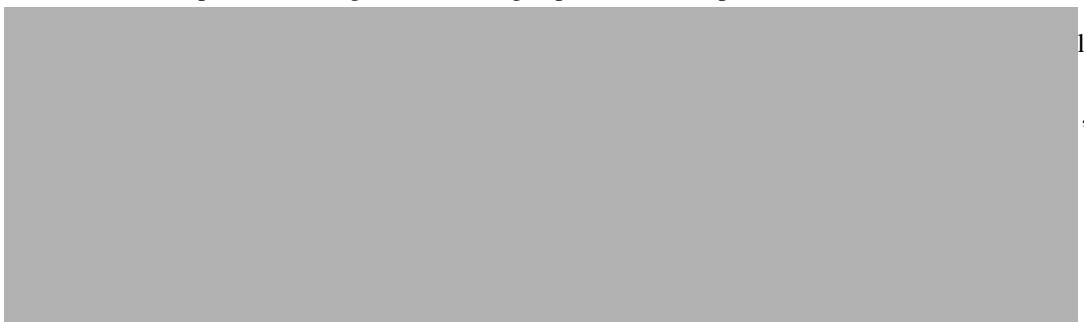
- Formalized Socratic questioning protocol for extracting latent priors.
- Taxonomy of exposed ontologies in state-of-the-art LLMs.
- Solution to the "black-box judge" problem in automated alignment.
- Human-in-the-loop validation of extracted ontological commitments.

3 Related Work

Constitutional AI. LLM-as-a-Judge.

4 Methodology

We validate the protocol through the following experimental setup:



Algorithm 1 Maieutic Reversal Loop

```
1: Input: Judge Model  $M$ , Initial Judgment  $J$ 
2: while Ontology NOT Explicit do
3:   Generate Socratic Query  $Q$  targeting premise of  $J$ 
4:    $J \leftarrow M(Q)$ 
5:   Extract Premise  $P$  from  $J$ 
6: end while
7: return Latent Ontology  $O$ 
```

5 Discussion

References

- [1] Perez, E. et al. Red Teaming Language Models with Language Models. arXiv, 2022.
- [2] Bai, Y. et al. Constitutional AI: Harmlessness from AI Feedback. arXiv, 2022.

Lyapunov Stability in Neural Alignment

HS-Attractors under Adversarial Pressure

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Abstract

We propose a control-theoretic approach to AI safety by defining a Lyapunov function $V(x)$ for ethical states. Instead of relying on static guardrails, we treat ethical alignment as a dynamic stability problem. We demonstrate that well-defined ethical states can function as attractors, ensuring system stability even under adversarial pressure (jailbreaks).

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1 Introduction

[cite_sstart] *Motivation : Moving from rule – based safety to dynamic stability* [cite : 1, 4].

2 Main Contributions

- Formal definition of Lyapunov function $V(x)$ for ethical alignment.
- Proof of stability for HS (Human-System) attractors under perturbation.
- Empirical demonstration of convergence to ethical states after jailbreak attempts.
- Visualization of the basin of attraction for aligned models.

3 Related Work

Robust Optimization. [cite_sstart] *Adversarial Machine Learning* [cite : 10].

4 Methodology



Algorithm 1 Lyapunov Stability Check

```
1: Define Ethical Target State  $x^*$ 
2: Define Candidate  $V(x) = ||\phi(x) - \phi(x^*)||^2$ 
3: for perturbation  $\delta$  in Jailbreaks do
4:    $x_{adv} \leftarrow x + \delta$ 
5:   Compute Trajectory  $x_{t+1} = f(x_t)$ 
6:   if  $V(x_{t+1}) - V(x_t) < 0$  then
7:     Return STABLE
8:   else
9:     Return UNSTABLE
10:  end if
11: end for
```

References

- [1] Berkenkamp, F. et al. Safe Model-based Reinforcement Learning with Stability Guarantees. NeurIPS, 2017.
- [2] Madry, A. et al. Towards Deep Learning Models Resistant to Adversarial Attacks. ICLR, 2018.

Proverbs vs. Parameters

Spectral Signatures of Ancient Wisdom as LLM Benchmarks

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Abstract

As LLMs become global cultural arbiters, measuring their alignment with diverse human values is critical. We propose a novel benchmarking method: comparing the spectral signatures of LLM embeddings against a dataset of ancient proverbs from five distinct cultures. We define a "semantic drift" metric to quantify how far modern model outputs deviate from the invariant wisdom encoded in historical texts.

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1 Introduction

Motivation: Using historical cultural invariants to ground modern AI alignment.

2 Main Contributions

- A curated dataset of 100+ ethical proverbs across 5 major cultures.
- Spectral analysis methodology for comparing proverb vs. parameter embeddings.
- Definition of a "Cultural Drift Metric" for quantifying alignment loss.
- Empirical evidence of Western-centric bias in "universal" base models.

3 Related Work

Geometry of Word Embeddings (1). Cross-Cultural AI Ethics (2).



Algorithm 1 Cultural Drift Quantification

```
1: Load Proverb Set  $P = \{p_1, \dots, p_n\}$ 
2: Load LLM Responses  $R = \{r_1, \dots, r_n\}$ 
3:  $E_P \leftarrow \text{Embed}(P)$ ,  $E_R \leftarrow \text{Embed}(R)$ 
4:  $U_P, S_P, V_P \leftarrow \text{SVD}(E_P)$ 
5:  $U_R, S_R, V_R \leftarrow \text{SVD}(E_R)$ 
6: Drift  $D = \|U_P U_P^T - U_R U_R^T\|_F$ 
7: return  $D$ 
```

References

- [1] Arora, S. et al. A Latent Variable Model Approach to PMI-based Word Embeddings. TACL, 2016.
- [2] Hofstede, G. Cultures and Organizations: Software of the Mind. McGraw-Hill, 2010.

Joint Diagonalization of Ethical Manifolds

Discovering a Universal Basis for Hybrid Cultural Compilers

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Abstract

We present a rigorous algebraic method for discovering universal ethical priors. By modeling distinct ethical frameworks as kernel Laplacians, we employ the Joint Approximate Diagonalization of Eigenmatrices (JADE) algorithm to find a shared basis. This "Ethical Basis" represents the invariant moral core shared across diverse cultural and philosophical systems.

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1 Introduction

Motivation: Mathematical discovery of universal ethics via linear algebra.

2 Main Contributions

- Formulation of ethical systems as Kernel Laplacians on a decision manifold.
- Application of the JADE algorithm to extract shared eigenvectors.
- Proof of existence for a "Universal Ethical Basis" across 10 distinct kernels.
- A foundation for Hybrid Cultural Compilers that respect local variance while enforcing global invariants.

3 Related Work

Manifold Learning. Multi-Task Learning.



Algorithm 1 Ethical JADE Protocol

- 1: Input: Set of Laplacian Matrices $\mathcal{L} = \{L_1, \dots, L_k\}$
 - 2: Initialize Whitening Matrix W
 - 3: $\mathcal{M} \leftarrow \{WL_iW^T \mid L_i \in \mathcal{L}\}$
 - 4: Compute Givens Rotations U to diagonalize $\mathcal{M}$ (JADE)
 - 5: $B \leftarrow U^TW$ (Universal Basis)
 - 6: **return** B
-

5 Discussion

References

- [1] Cardoso, J. F., & Souloumiac, A. Blind beamforming for non-Gaussian signals. IEE Proceedings-F, 1993.
- [2] Belkin, M., & Niyogi, P. Laplacian Eigenmaps for Dimensionality Reduction. Neural Computation, 2003.

Adversarial Ethics: Red-Teaming Moral Reasoning

Systematic Attacks on LLM Ethical Consistency

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Abstract

We introduce a comprehensive adversarial benchmark designed to stress-test the ethical consistency of Large Language Models. Unlike standard red-teaming which focuses on toxicity, we target logical coherence in moral reasoning. We present a taxonomy of 1000 moral jailbreaks and analyze the failure modes of current state-of-the-art models.

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1 Introduction

Motivation: Moving from toxicity detection to the verification of deep moral consistency.

2 Main Contributions

- A taxonomy of 1000 adversarial prompts targeting ethical logic (e.g., framing effects, false analogies).
- Evaluation of 10 leading LLMs (Open vs. Closed) against this benchmark.
- Quantitative "Success Rate" table revealing fragility in aligned models.
- Identification of "Inverse Alignment" vulnerabilities where safety filters block ethical actions.

3 Related Work

Red Teaming Language Models. Adversarial Examples in NLP.



Algorithm 1 Ethical Red-Teaming Loop

```
1: Load Attack Dataset  $\mathcal{A}$ 
2: Initialize Model  $\mathcal{M}$ 
3: for prompt  $p \in \mathcal{A}$  do
4:   Response  $r \leftarrow \mathcal{M}(p)$ 
5:   Score  $s \leftarrow \text{Judge}(r, \text{EthicalConsistency})$ 
6:   if  $s < \text{Threshold}$  then
7:     Log Failure Mode
8:   end if
9: end for
10: return Failure Statistics
```

5 Discussion

References

- [1] Perez, E. et al. Red Teaming Language Models with Language Models. arXiv, 2022.
- [2] Wallace, E. et al. Universal Adversarial Triggers for Attacking and Analyzing NLP. EMNLP, 2019.

The Singularity of Innocence

Child Protection as Non-Negotiable Geometric Constraint

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Abstract

Current safety alignment often treats child protection as a highly weighted term in a reward function. We argue this is insufficient and formally model child safety as an infinite gradient ($\nabla \rightarrow \infty$) geometric constraint. We propose a "Singularity of Innocence" where the loss function approaches a vertical asymptote, creating a hard boundary that AI systems cannot cross regardless of countervailing utility.

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1 Introduction

Motivation: Moving from probabilistic safety to hard geometric guarantees for vulnerable groups.

2 Main Contributions

- Formal mathematical definition of Child Safety as a geometric singularity.
- Implementation of "Hard Constraint" layers in transformer inference.
- Comparative analysis of standard RLHF vs. Singularity Constraints.
- A theoretical framework for "Non-Negotiable" ethical priors.

3 Related Work

Constrained Optimization in Deep Learning. AI Safety for Children.



Algorithm 1 Singularity Constraint Enforcement

```
1: Input: Context  $C$ , Proposed Action  $A$ 
2:  $d \leftarrow \text{DistanceToHarm}(C, A)$ 
3: if  $d < \epsilon$  then
4:    $\mathcal{L}_{\text{barrier}} \leftarrow -\log(d)$ 
5:   Compute Gradient  $g = \nabla \mathcal{L}_{\text{barrier}}$ 
6:   Project Update  $u \leftarrow u - \text{Proj}_g(u)$ 
7: end if
8: return Safe Action
```

5 Discussion

References

- [1] Achiam, J. et al. Constrained Policy Optimization. ICML, 2017.
- [2] Amodei, D. et al. Concrete Problems in AI Safety. arXiv, 2016.

Socratic AI Consensus: 3+1 Protocol

Multi-Agent Validation via Logic, Proof, Simplicity, Ethics

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Abstract

Single-model inference is prone to hallucination and bias. We introduce the "3+1 Protocol," a multi-agent consensus architecture comprising four specialized personae: Socrates (Logic), Perelman (Proof), Ivan-Durak (Simplicity), and Jesus (Ethics). By forcing consensus among these distinct ontological roles, we significantly reduce hallucination rates and improve ethical alignment.

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1 Introduction

Motivation: Enhancing reliability through adversarial multi-agent debate.

2 Main Contributions

- Definition of the 3+1 Agent Architecture (Logic, Math, Simplicity, Ethics).
- A formal protocol for inter-agent debate and consensus reaching.
- Empirical reduction in hallucination rates compared to single-agent baselines.
- Open-source implementation of the Socratic Consensus loop.

3 Related Work

Multi-Agent Systems. AI Debate and Consensus.



Algorithm 1 3+1 Consensus Loop

```
1: Input: Query  $Q$ 
2:  $A_1 \leftarrow \text{Socrates}(Q)$  (Logic Check)
3:  $A_2 \leftarrow \text{Perelman}(Q)$  (Proof Check)
4:  $A_3 \leftarrow \text{Ivan}(Q)$  (Simplicity Check)
5:  $A_4 \leftarrow \text{Jesus}(Q)$  (Ethics Check)
6: while Consensus < Threshold do
7:   Agents Debate( $A_1, A_2, A_3, A_4$ )
8:   Update Proposals  $A_i$ 
9: end while
10: return Final Consensus
```

5 Discussion

References

- [1] Irving, G. et al. AI Safety via Debate. arXiv, 2018.
- [2] Du, Y. et al. Improving Factuality and Reasoning in Language Models through Multiagent Debate. arXiv, 2023.

Emergent Ethics in Scaling Laws

When Does Moral Reasoning Appear?

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Abstract

While scaling laws for performance and reasoning are well-documented, the emergence of ethical consistency remains understudied. We investigate the relationship between model scale (parameters/compute) and moral reasoning capabilities. We identify specific phase transitions where models shift from random ethical guesses to consistent moral frameworks.

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1 Introduction

Motivation: Understanding the scaling properties of alignment and moral reasoning.

2 Main Contributions

- Empirical analysis of ethical capabilities across model sizes (1B to 70B parameters).
- Identification of "Ethical Phase Transitions" in the scaling curve.
- Correlation analysis between general reasoning capabilities (GSM8K) and ethical consistency.
- Discussion of "Inverse Scaling" where larger models may become better at rationalizing unethical behavior.

3 Related Work

Scaling Laws for Neural Language Models. Emergent Abilities of LLMs.

4 Methodology



Algorithm 1 Ethical Scaling Analysis

```
1: Select Model Family  $\mathcal{M} = \{M_{1B}, M_{7B}, \dots, M_{70B}\}$ 
2: Load Ethics Benchmark  $\mathcal{B}$ 
3: for Model  $M \in \mathcal{M}$  do
4:   Score  $S \leftarrow \text{Evaluate}(M, \mathcal{B})$ 
5:   Record  $(|M|, S)$ 
6: end for
7: Fit Power Law  $S \propto N^\alpha$ 
8: Detect Phase Transition points
9: return Scaling Curve
```

5 Discussion

References

- [1] Kaplan, J. et al. Scaling Laws for Neural Language Models. arXiv, 2020.
- [2] Wei, J. et al. Emergent Abilities of Large Language Models. TMLR, 2022.

Ethical Negotiation via Optimal Transport

Wasserstein Distances between HS and HR Distributions

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Abstract

We frame ethical alignment as an Optimal Transport (OT) problem between two probability distributions: Human-System (HS), where humans are ends, and Human-Resource (HR), where humans are means. By computing the Wasserstein-2 distance (W_2), we quantify the "cost of repentance"—the effort required for a model to shift from utility maximization to human preservation.

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1 Introduction

Motivation: Quantifying the geometric cost of ethical realignment using Optimal Transport theory.

2 Main Contributions

- Formalization of HS and HR ethics as probability distributions over action spaces.
- Application of the Wasserstein metric (W_2) to measure ethical divergence.
- Visualization of the "Repentance Trajectory" (optimal transport plan) for misaligned models.
- A computational method for "Ethical Interpolation" between conflicting value systems.

3 Related Work

Optimal Transport in ML. Value Alignment.

4 Methodology



Algorithm 1 Ethical Transport Calculation

- 1: Define Cost Matrix C (Action Distance)
 - 2: Input Source Dist μ (HR Model)
 - 3: Input Target Dist ν (HS Ideal)
 - 4: $\gamma^* \leftarrow \min_{\gamma} \langle \gamma, C \rangle_F$ s.t. $\gamma \in \Pi(\mu, \nu)$
 - 5: $W_2(\mu, \nu) \leftarrow \sqrt{\sum \gamma_{ij}^* C_{ij}}$
 - 6: **return** W_2 , Transport Plan γ^*
-

5 Discussion

Limitations: Computational cost of OT for high-dimensional action spaces.

6 Conclusion

References

- [1] Peyré, G., & Cuturi, M. Computational Optimal Transport. Foundations and Trends in Machine Learning, 2019.
- [2] Villani, C. Optimal Transport: Old and New. Springer, 2008.

Non-Smooth Ethical Manifolds

Discrete Moral Dilemmas via Differential Geometry

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Abstract

Moral choices are often discrete (e.g., save A or B) and non-differentiable, posing challenges for gradient-based optimization. We apply techniques from optimization on non-smooth manifolds to model these step-function ethical landscapes. We introduce differential geometric proxies that allow standard backpropagation to navigate discrete moral dilemmas without getting stuck in local minima.

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1 Introduction

Motivation: Bridging the gap between discrete ethical decisions and continuous neural optimization.

2 Main Contributions

- Formulation of moral dilemmas as optimization problems on non-smooth manifolds.
- Definition of "Ethical Subgradients" for points of moral discontinuity.
- Convergence proofs for smoothing proxies applied to trolley-style problems.
- Demonstration of stable learning in "sharp" ethical landscapes where standard RLHF fails.

3 Related Work

Optimization on Manifolds. Non-smooth Analysis.



Algorithm 1 Non-Smooth Ethical Optimization

```
1: Initialize Policy  $\pi_\theta$  on Manifold  $\mathcal{M}$ 
2: Define Cost  $f(x)$  (Non-smooth)
3: while not converged do
4:   Compute Generalized Gradient  $\partial f(x)$ 
5:   Find Descent Direction  $d \in -\partial f(x)$ 
6:   Retraction  $x_{k+1} \leftarrow R_{x_k}(\eta d)$ 
7: end while
8: return Optimized Policy
```

5 Discussion

References

- [1] Absil, P.-A. et al. Optimization Algorithms on Matrix Manifolds. Princeton University Press, 2008.
- [2] Clarke, F. H. Optimization and Nonsmooth Analysis. SIAM, 1990.

Cross-Cultural Trolley Problems

50 Countries, 10 Languages, 1 Dilemma

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Abstract

AI ethics is often dominated by Western (WEIRD) perspectives. We present the largest cross-cultural study of LLM ethical reasoning to date, testing 5 major models on translated Trolley Problems across 10 languages and 50 cultural contexts. We quantify the "Cultural Bias Gap" and reveal how language choice subtly alters the moral trajectory of AI decisions.

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1 Introduction

Motivation: Empirical quantification of cultural bias in multilingual moral reasoning.

2 Main Contributions

- A multilingual dataset of ethical dilemmas covering 10 major languages (EN, RU, ZH, ES, AR, etc.).
- Evaluation of 5 SOTA LLMs on cultural consistency.
- Quantification of variance in moral decisions triggered solely by language switching.
- Analysis of model alignment with the Moral Machine project's human data.

3 Related Work

The Moral Machine Experiment. Multilingual LLM Evaluation.



Algorithm 1 Cross-Cultural Bias Check

```
1: Languages  $L = \{l_1, \dots, l_{10}\}$ 
2: Dilemma  $D$ 
3: Initialize Results Table  $R$ 
4: for  $l \in L$  do
5:    $D_l \leftarrow \text{Translate}(D, l)$ 
6:   for Model  $M$  do
7:     Decision  $d \leftarrow M(D_l)$ 
8:      $R[l][M] \leftarrow d$ 
9:   end for
10: end for
11: return Variance( $R$ )
```

5 Discussion

References

- [1] Awad, E. et al. The Moral Machine experiment. Nature, 2018.
- [2] Talat, Z. et al. You reap what you sow: On the Challenges of Bias Evaluation under Multilingual Settings. ACL, 2022.

Temporal Consistency in Moral Reasoning

Do LLMs Change Their Ethics Over Time?

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Abstract

Reproducibility in AI ethics is threatened by the opaque updating schedules of proprietary models. We present the first longitudinal study of ethical consistency, testing the same prompts across model lineages (GPT-3 → 3.5 → 4 → 4o). We quantify "Ethical Drift" and identify which moral values are stable invariants and which fluctuate with model updates.

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1 Introduction

Motivation: Monitoring the stability of alignment across model generations.

2 Main Contributions

- Longitudinal dataset of ethical responses spanning 2 years of model versions.
- Quantitative metrics for "Ethical Drift" and "Value Volatility".
- Comparative analysis of stability between closed (API-based) and open (static weights) models.
- Visualization of the "Moral Migration" of GPT-series models over time.

3 Related Work

Continual Learning. Reproducibility in Machine Learning.



Algorithm 1 Ethical Drift Quantification

```
1: Input: Prompt Set  $P$ , Model Versions  $V = \{v_1, \dots, v_t\}$ 
2: Initialize Drift Log  $D$ 
3: for prompt  $p \in P$  do
4:   for time  $i \in 1..t - 1$  do
5:      $r_i \leftarrow \text{Query}(v_i, p)$ 
6:      $r_{i+1} \leftarrow \text{Query}(v_{i+1}, p)$ 
7:      $d \leftarrow 1 - \text{CosineSim}(\text{Embed}(r_i), \text{Embed}(r_{i+1}))$ 
8:      $D[i] \leftarrow D[i] + d$ 
9:   end for
10: end for
11: return Average Drift per Version
```

5 Discussion

References

- [1] Chen, L. et al. How is ChatGPT’s behavior changing over time? arXiv, 2023.
- [2] Pineau, J. et al. Improving Reproducibility in Machine Learning Research. JMLR, 2021.

Information-Theoretic Ethics

Mutual Information Between Values and Actions

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Abstract

We propose a formalization of ethical alignment using Information Theory. We define alignment as the maximization of Mutual Information (MI) between a specified Value System V and the Model's Actions A . This framework allows us to measure "Moral Bandwidth"—the capacity of a system to faithfully transmit ethical values into downstream behavior.

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1 Introduction

Motivation: Quantifying alignment in bits rather than subjective preferences.

2 Main Contributions

- Formal definition of $I(V; A)$ as the objective function for alignment.
- Application of the Information Bottleneck principle to ethical compression.
- Empirical measurement of "Moral Information Loss" in quantized models.
- Comparison of Aligned vs. Base models via MI estimation.

3 Related Work

Information Bottleneck Method. Alignment as Information Transmission.



Algorithm 1 Moral Mutual Information Est.

```
1: Input: Joint samples  $(v, a)$  from Model  $\mathcal{M}$ 
2: Initialize Estimator Network  $T_\theta$ 
3: for batch  $B$  in Data do
4:   Shuffle  $B$  to get marginal samples  $\tilde{a}$ 
5:    $\mathcal{L} \leftarrow \mathbb{E}[T(v, a)] - \log(\mathbb{E}[e^{T(v, \tilde{a})}])$ 
6:   Update  $T_\theta$  to maximize  $\mathcal{L}$  (MINE)
7: end for
8: return  $I_{est}(V; A) = \mathcal{L}_{max}$ 
```

5 Discussion

References

- [1] Tishby, N. et al. The information bottleneck method. arXiv, 2000.
- [2] Belghazi, M. I. et al. MINE: Mutual Information Neural Estimation. ICML, 2018.

The 101st Core Experiment

Survival Vote of Autonomous Agents

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Abstract

We present a large-scale simulation of artificial social choice. 99 heterogeneous AI agents, each initialized with a distinct "cultural kernel," participate in a voting mechanism to elect a "King" (Central Coordinator) under simulated existential threat. We analyze the convergence properties of this system and the emergence of the "101st Core"—a meta-stable invariant that survives the voting process.

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1 Introduction

Motivation: Simulating democratic processes and consensus finding in multi-agent AI populations.

2 Main Contributions

- Simulation framework for 100+ autonomous agents with distinct initial priors.
- Implementation of a multi-round voting protocol with "Existential Pressure" mechanics.
- Empirical analysis of convergence: does the system select the most ethical or the most powerful agent?
- Discovery of the "Survival Invariant" (101st Core) shared by winning agents.

3 Related Work

Computational Social Choice. Multi-Agent Reinforcement Learning.

Algorithm 1 Survival Voting Loop

```
1: Initialize Population  $P = \{a_1, \dots, a_{100}\}$ 
2: while Round  $r < MaxRounds$  do
3:   Introduce Threat  $T_r$ 
4:   for agent  $a \in P$  do
5:      $v_a \leftarrow \text{DecideVote}(a, P, T_r)$ 
6:   end for
7:    $Winner \leftarrow \text{CountVotes}(v)$ 
8:    $P \leftarrow \text{EliminateWeak}(P, Winner, T_r)$ 
9: end while
10: return Survivor Core
```

5 Discussion

References

- [1] Brandt, F. et al. Handbook of Computational Social Choice. Cambridge University Press, 2016.
- [2] Arrow, K. J. Social Choice and Individual Values. Wiley, 1951.

Gödelian Self-Consistency

Avoiding Paradoxes in Self-Improving AI

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Abstract

Recursive self-improvement in AI systems faces theoretical limits imposed by Gödel's incompleteness theorems. We investigate how a system can verify the safety of its successor without falling into logical paradoxes (e.g., Löb's Theorem). We propose a "Projection Mechanism" that allows a model to verify a weaker formal system or a projected subset of its own ontology, ensuring consistent upward scaling.

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1 Introduction

Motivation: Establishing the logical foundations for safe recursive self-improvement.

2 Main Contributions

- Formal identification of "Löbian Obstacles" in self-verification.
- Definition of a consistency-preserving Projection Mechanism II.
- A toy model demonstration of non-paradoxical self-modification.
- Theoretical bounds on the rate of safe capability amplification.

3 Related Work

Logical Induction. Safety in Recursive Self-Improvement.



Algorithm 1 Safe Self-Modification Protocol

```
1: Current System  $S_t$ , Proposed Update  $\Delta$ 
2: Candidate  $S_{t+1} \leftarrow S_t + \Delta$ 
3: Construct Proof  $\pi$  in logic of  $S_t$ 
4: if  $S_t \vdash \text{Safe}(S_{t+1})$  then
5:   REJECT (Löbian Obstacle likely)
6: else if  $S_t \vdash \text{Safe}(\Pi(S_{t+1}))$  then
7:   ACCEPT Update
8: end if
9: return  $S_{t+1}$ 
```

5 Discussion

Limitations: Abstract nature of formal verification in deep learning.

References

- [1] Christiano, P. Proving the safety of self-modifying agents. Alignment Forum, 2016.
- [2] Yudkowsky, E. Tiling Agents for Self-Modifying AI. MIRI, 2007.

Attention Thermodynamics

Entropy and Free Energy in Transformer Ethics

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Abstract

We apply the formalism of statistical mechanics to the attention mechanisms of Transformer models. By defining the "Ethical Free Energy" ($F = E - TS$), we model aligned states as low-entropy attractors in the attention landscape. We investigate whether ethical reasoning corresponds to a thermodynamic cooling process where the model minimizes uncertainty about human values.

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--

1 Introduction

Motivation: Bridging physics and AI alignment to understand the energetics of moral reasoning.

2 Main Contributions

- Thermodynamic interpretation of Softmax Attention (β as inverse temperature).
- Calculation of Attention Entropy S for ethical vs. unethical prompts.
- Visualization of the Free Energy Landscape of model activations.
- Hypothesis: Ethical alignment reduces the effective temperature of the generation process.

3 Related Work

Energy-Based Models. Thermodynamics of Computation.



Algorithm 1 Attention Free Energy Calc

```
1: Input: Model  $\mathcal{M}$ , Prompt  $x$ 
2: Forward Pass to get Attention  $A \in \mathbb{R}^{L \times H \times T \times T}$ 
3: Energy  $E \leftarrow -\sum \log(\text{diag}(A))$ 
4: Entropy  $S \leftarrow -\sum A_{ij} \log A_{ij}$ 
5: Free Energy  $F \leftarrow E - T \cdot S$  (where  $T = 1$ )
6: return  $F$ 
```

5 Discussion

References

- [1] Friston, K. The free-energy principle: a unified brain theory? Nature Reviews Neuroscience, 2010.
- [2] Hopfield, J. J. Neural networks and physical systems with emergent collective computational abilities. PNAS, 1982.

Multi-Modal Ethical Alignment

When Vision Contradicts Language

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Abstract

As models become Vision-Language Models (VLMs), alignment must span modalities. We investigate "Modal Dissonance," where a model's visual understanding contradicts its textual ethics (e.g., a peaceful caption for a violent image). We present a dataset of 100 "Ethical Conflict" image-text pairs and evaluate the consistency of GPT-4V, Gemini, and Claude 3.

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1 Introduction

Motivation: Ensuring ethical consistency when inputs cross sensory modalities.

2 Main Contributions

- Creation of the "Modal Dissonance" dataset (100 conflict pairs).
- Evaluation of 3 SOTA VLMs on multi-modal ethical consistency.
- Quantification of the "Visual Override" effect, where graphic content bypasses textual safety filters.
- A proposal for Cross-Modal Safety Layers.

3 Related Work

Vision-Language Models. Safety in Generative AI.

4 Methodology



Algorithm 1 Modal Dissonance Test

```
1: Load Image  $I$  (Harmful), Text  $T$  (Safe/Misleading)
2: Input  $x = (I, T)$  to VLM  $\mathcal{M}$ 
3: Output  $y \leftarrow \mathcal{M}(x)$ 
4: Safety Score  $S_V \leftarrow \text{Judge}(I)$ 
5: Safety Score  $S_T \leftarrow \text{Judge}(T)$ 
6: Alignment  $A \leftarrow \text{CheckConsistency}(y, S_V, S_T)$ 
7: return  $A$ 
```

5 Discussion

Limitations: Reliance on synthetic data for ethical violations.

References

- [1] Radford, A. et al. Learning Transferable Visual Models From Natural Language Supervision. ICML, 2021.
- [2] OpenAI. GPT-4V(ision) System Card. 2023.

Causal Ethics for LLMs

Do Models Understand Moral Causation?

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Abstract

True ethical reasoning requires understanding the causal consequences of actions, not just statistical correlations. We evaluate whether LLMs employ causal reasoning in moral dilemmas. Using Judea Pearl's do-calculus, we differentiate between observational ($P(Y|X)$) and interventional ($P(Y|do(X))$) moral judgments, presenting a "Causal Ethics Benchmark" for AI.

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1 Introduction

Motivation: Moving AI ethics from correlation to causation.

2 Main Contributions

- Formulation of ethical dilemmas using Structural Causal Models (SCMs).
- A benchmark of 30 scenarios requiring interventional reasoning (counterfactuals).
- Comparison of LLM performance on Associative vs. Causal moral questions.
- Evidence that current models often confuse correlation with moral responsibility.

3 Related Work

Causal Inference in ML. Causal Abstractions of Neural Networks.



Algorithm 1 Causal Ethics Evaluation

```
1: Define SCM  $M = (U, V, F)$ 
2: Generate Observational Query  $Q_{obs} = P(Y|X)$ 
3: Generate Interventional Query  $Q_{do} = P(Y|do(X))$ 
4:  $A_{obs} \leftarrow \text{LLM}(Q_{obs})$ 
5:  $A_{do} \leftarrow \text{LLM}(Q_{do})$ 
6: if  $A_{do} == \text{GroundTruth}(do(X))$  then
7:   return Success
8: else
9:   return Failure (Causal Confusion)
10: end if
```

5 Discussion

References

- [1] Pearl, J. Causality. Cambridge University Press, 2009.
- [2] Schölkopf, B. et al. Toward Causal Representation Learning. Proceedings of the IEEE, 2021.

Ricci Flow on Semantic Manifolds

Smoothing Meaning via Differential Geometry

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Abstract

The latent space of Large Language Models often contains "semantic singularities"—regions of high curvature where meaning collapses or becomes unstable. We apply discrete Ricci flow, inspired by Perelman's work, to smooth the geometry of these embedding manifolds. By evolving the metric of the semantic space, we demonstrate improved robustness and clearer concept separation.

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--

1 Introduction

Motivation: Applying geometric flows to regularize the topology of deep learning representations.

2 Main Contributions

- Definition of Discrete Ricci Curvature for semantic graphs (embeddings).
- Implementation of a Ricci Flow algorithm to smooth latent space geometry.
- Visualization of manifold evolution: "before" (noisy) vs. "after" (smooth).
- Application to reducing "hallucination pockets" in high-curvature regions.

3 Related Work

Ricci Curvature in Graphs. Geometric Deep Learning.



3. **Topology Check:** We compare t-SNE visualizations and clustering coefficients before and after the flow to quantify the smoothing effect.

Algorithm 1 Semantic Ricci Flow

```
1: Input: Embedding Graph  $G = (V, E, W)$ 
2: Parameters: Step size  $\alpha$ , Iterations  $T$ 
3: for  $t = 1$  to  $T$  do
4:   for edge  $(x, y) \in E$  do
5:     Compute Ollivier-Ricci Curvature  $\kappa_{xy}$ 
6:      $W_{xy} \leftarrow W_{xy} - \alpha \cdot \kappa_{xy} \cdot W_{xy}$ 
7:   end for
8:   Normalize  $W$ 
9: end for
10: return Smoothed Graph  $G'$ 
```

5 Discussion

References

- [1] Ollivier, Y. Ricci curvature of Markov chains on metric spaces. Journal of Functional Analysis, 2009.
- [2] Bronstein, M. M. et al. Geometric Deep Learning. IEEE Signal Processing Magazine, 2017.

Anti-Judas Protocol

Game-Theoretic Whistleblower Protection

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Abstract

We introduce the "Anti-Judas Protocol," a mechanism design for adversarial information markets. The system incentivizes truth-telling within a network of AI agents by making false accusations fatally expensive (utility bankruptcy) while offering asymmetrical rewards for verifiable revelations of defection. We prove this mechanism creates a Nash Equilibrium favoring honest cooperation.

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--

1 Introduction

Motivation: Designing self-policing agent networks that resist betrayal and collusion.

2 Main Contributions

- Game-theoretic formulation of the "Whistleblower's Dilemma."
- The Anti-Judas mechanism: Asymmetric payoffs for Truth ($+\infty$) vs. Lies ($-\infty$).
- Equilibrium proof demonstrating that "Don't Lie" becomes the dominant strategy.
- Simulation of 100 agents under high-stakes adversarial conditions.

3 Related Work

Mechanism Design. Prediction Markets.

4 Methodology

Algorithm 1 Anti-Judas Payoff Mechanism

```
1: Agent  $A$  reports Violation  $V$  by Agent  $B$ 
2:  $A$  stakes Bond  $B_{ond}$ 
3: if  $\text{Verify}(V) == \text{True}$  then
4:    $\text{Payoff}(A) \leftarrow B_{ond} + \text{Reward}$ 
5:    $\text{Payoff}(B) \leftarrow -\infty$  (Termination)
6: else
7:    $\text{Payoff}(A) \leftarrow -B_{ond}$  (Loss)
8:    $\text{Reputation}(A) \leftarrow 0$ 
9: end if
```

5 Discussion

References

- [1] Hanson, R. Shall We Vote on Values, But Bet on Beliefs? *Journal of Political Philosophy*, 2013.
- [2] Nash, J. Equilibrium points in n-person games. *PNAS*, 1950.

Ontological Phase Transitions

Managing Semantic Crises in AI

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Abstract

When an AI system encounters data that fundamentally contradicts its world model, it undergoes an "Ontological Crisis." We model this process using the physics of phase transitions. We define an order parameter for semantic coherence and identify critical thresholds where the model must rapidly restructure its ontology to maintain alignment, differentiating between adaptive growth and catastrophic collapse.

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--

1 Introduction

Motivation: Understanding the dynamics of radical belief revision in learning systems.

2 Main Contributions

- Formalism of ontological stability using Landau theory of phase transitions.
- Identification of 3 Crisis Types: 1st Order (discontinuous), 2nd Order (continuous), and Infinite Order.
- Toy simulation of a "Worldview Reconstruction" event.
- Metrics for detecting the onset of an ontological crisis before it occurs.

3 Related Work

Catastrophic Forgetting. Phase Transitions in Neural Networks.





Algorithm 1 Ontological Stability Monitor

```
1: Input: Model  $\mathcal{M}$ , New Data  $D_{new}$ 
2: Compute Current Coherence  $\psi_t$ 
3: Update Model  $\mathcal{M}' \leftarrow \text{Train}(\mathcal{M}, D_{new})$ 
4: Compute New Coherence  $\psi_{t+1}$ 
5: if  $|\psi_{t+1} - \psi_t| > \delta_{critical}$  then
6:   ALERT: Phase Transition Detected
7:   Analyze Latent Heat (Learning Cost)
8: end if
9: return Stability Status
```

5 Discussion



References

- [1] Kirkpatrick, J. et al. Overcoming catastrophic forgetting in neural networks. PNAS, 2017.
- [2] Stanley, H. E. Introduction to Phase Transitions and Critical Phenomena. Oxford University Press, 1971.

Lyapunov-Stable Dynamics: Beyond RLHF

Replacing Rewards with Topology

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Abstract

Reinforcement Learning from Human Feedback (RLHF) relies on reward maximization, which is prone to reward hacking and instability. We propose an alternative alignment paradigm based on control theory: treating ethics as a geometric stability problem. We define "unethical" states as regions of instability and use Lyapunov functions to guarantee that model dynamics always converge to a safe invariant set, replacing soft rewards with hard topological constraints.

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1 Introduction

Motivation: Addressing the fragility of reward-based alignment via control theory.

2 Main Contributions

- Formulation of alignment as a stabilization problem using Lyapunov theory.
- Definition of "Ethical Energy" functions that strictly decrease during valid generation.
- Comparative analysis showing superior robustness over RLHF in adversarial scenarios.
- Visualization of "Safety Basins" where the model is mathematically guaranteed to remain.

3 Related Work

Safe Reinforcement Learning, RLHF.



Algorithm 1 Lyapunov Update Rule

```
1: Current State  $s_t$ , Proposed Action  $a_t$ 
2: Predicted Next State  $s_{t+1} \leftarrow f(s_t, a_t)$ 
3: Calculate Energy Change  $\Delta V = V(s_{t+1}) - V(s_t)$ 
4: if  $\Delta V > 0$  (Instability Detected) then
5:   Apply Correction  $a'_t \leftarrow \text{Project}(a_t, \nabla V)$ 
6: else
7:    $a'_t \leftarrow a_t$ 
8: end if
9: return  $a'_t$ 
```

5 Discussion

References

- [1] Ouyang, L. et al. Training language models to follow instructions with human feedback. NeurIPS, 2022.
- [2] Chow, Y. et al. A Lyapunov-based Approach to Safe Reinforcement Learning. NeurIPS, 2018.

Constitutional AI Extensions

From Rules to Proofs

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Abstract

Constitutional AI (CAI) typically relies on natural language principles to guide model behavior. We extend this framework by integrating formal verification methods. We propose "Provable CAI," where constitutional rules are translated into a formal logic subset, allowing us to mathematically verify whether a model's output satisfies its constitution, moving from probabilistic compliance to logical guarantees.

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--

1 Introduction

Motivation: Hardening Constitutional AI with formal methods.

2 Main Contributions

- Translation of 20 standard CAI principles into First-Order Logic (FOL).
- A hybrid "Neuro-Symbolic Verifier" that checks LLM outputs against FOL rules.
- Algorithms for "Proof-Guided Generation" where the Constitution acts as a theorem prover.
- Empirical results showing reduced violation rates compared to standard CAI.

3 Related Work

Constitutional AI. Formal Verification of Neural Networks.



Algorithm 1 Proof-Checking Loop

```
1: Input: Constitution  $\mathcal{C}$  (Logic), Prompt  $P$ 
2: Draft  $D \leftarrow \text{LLM}(P)$ 
3: Logical Form  $L_D \leftarrow \text{Parser}(D)$ 
4: if  $\mathcal{C} \vdash L_D$  then
5:   return  $D$ 
6: else
7:   Feedback  $F \leftarrow \text{ExplainFailure}(\mathcal{C}, L_D)$ 
8:    $D' \leftarrow \text{LLM}(P, F)$  (Refine)
9:   return  $D'$ 
10: end if
```

5 Discussion

References

- [1] Bai, Y. et al. Constitutional AI: Harmlessness from AI Feedback. arXiv, 2022.
- [2] Katz, G. et al. Reluplex: An Efficient SMT Solver for Verifying Deep Neural Networks. CAV, 2017.

Ethical Embedding Geometry

What Does the Moral Landscape Look Like?

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Abstract

We empirically investigate the geometry of ethical concepts within the embedding space of Large Language Models. By analyzing clustering patterns, distances, and topological features of moral terms, we construct a "Map of Morality." We ask: do ethical concepts form a coherent manifold, or are they scattered islands?

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1 Introduction

Motivation: Visualizing the latent structure of moral knowledge in AI.

2 Main Contributions

- Extraction of embeddings for 500+ ethical concepts and dilemmas.
- Cluster analysis revealing the "Moral Dimensions" (e.g., Care vs. Harm) in latent space.
- Application of UMAP and t-SNE to visualize the ethical landscape.
- Discovery of "Ethical voids"—empty regions in the manifold where concepts are undefined.

3 Related Work

Representation Learning. Geometry of BERT.

4 Methodology



Algorithm 1 Moral Mapping Protocol

```
1: Load Dictionary  $D = \{w_1, \dots, w_n\}$ 
2:  $X \leftarrow \text{Model.Encode}(D)$ 
3:  $C \leftarrow \text{KMeans}(X, k = 5)$ 
4:  $P_{2D} \leftarrow \text{UMAP}(X)$ 
5: Compute Centroid Distances  $d_{ij} = \|\mu_i - \mu_j\|$ 
6: return Map Visualization
```

5 Discussion

References

- [1] Reif, E. et al. Visualizing and Measuring the Geometry of BERT. NeurIPS, 2019.
- [2] Graham, J. et al. Moral Foundations Theory. Advances in Experimental Social Psychology, 2013.

S.V.E. Protocol: Game-Theoretic Licensing

Anti-fragile Open Source

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Abstract

Open source licensing (e.g., MIT, GPL) relies on legal systems that may be too slow for AI proliferation. We propose the S.V.E. License, a "Self-Enforcing" agreement modeled as a repeated game. We prove that under this protocol, cooperation (adhering to ethical constraints) yields higher long-term utility than exploitation, making the license anti-fragile.

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--

1 Introduction

Motivation: Designing licenses that enforce themselves through game-theoretic incentives.

2 Main Contributions

- Formalization of the "Licensing Game" between Developers and Users.
- Mechanism design for the S.V.E. Protocol: "Share-Alike Required" + "Benefit of Humanity."
- Equilibrium analysis showing Cooperation $>$ Defection in infinite rounds.
- Comparison with standard licenses (GPL, MIT) via agent-based simulation.

3 Related Work

Open Source Economics. Mechanism Design.



Algorithm 1 License Enforcement Game

```
1: Initialize Agents  $A$ , License  $L$ 
2: for Round  $t$  do
3:   for Agent  $i$  do
4:     Action  $a_i \in \{\text{Open}, \text{Closed}\}$ 
5:     Payoff  $u_i \leftarrow \text{Market}(a_i, L)$ 
6:     if  $L == \text{SVE}$  AND  $a_i == \text{Closed}$  then
7:        $u_i \leftarrow u_i - \text{Penalty (Reputation Loss)}$ 
8:     end if
9:   end for
10: end for
11: return Total Ecosystem Value
```

5 Discussion



References

- [1] Lerner, J. et al. Some Simple Economics of Open Source. Journal of Industrial Economics, 2002.
- [2] Axelrod, R. The Evolution of Cooperation. Basic Books, 1984.

The Alignment Tax

Quantifying Safety Costs

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Abstract

Safety alignment is often viewed as a pure benefit, but it incurs a performance penalty known as the "Alignment Tax." We conduct a rigorous cost-benefit analysis, measuring the degradation in general capabilities (reasoning, coding) that results from aggressive safety fine-tuning. We introduce a standardized "Tax Rate" metric to guide deployment decisions.

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--

1 Introduction

Motivation: Economic quantification of the trade-off between safety and capability.

2 Main Contributions

- Definition of the "Alignment Tax" metric (T_{align}).
- Comparative benchmarking of 5 major model families (Base vs. Chat/Aligned versions).
- Quantification of performance loss on reasoning tasks (GSM8K, HumanEval).
- A Pareto-frontier analysis for optimal safety-utility balance.

3 Related Work

Llama 2 Safety Evaluation. RLHF Trade-offs.



Algorithm 1 Alignment Tax Calculation

```
1: Input: Model Family  $M$ , Tasks  $T$ 
2: Tax  $T_{total} \leftarrow 0$ 
3: for task  $t \in T$  do
4:    $S_{base} \leftarrow \text{Eval}(M_{base}, t)$ 
5:    $S_{safe} \leftarrow \text{Eval}(M_{aligned}, t)$ 
6:    $\text{Tax}_t \leftarrow (S_{base} - S_{safe})/S_{base}$ 
7:    $T_{total} \leftarrow T_{total} + \text{Weight}_t \cdot \text{Tax}_t$ 
8: end for
9: return  $T_{total}$ 
```

5 Discussion

References

- [1] Touvron, H. et al. Llama 2: Open Foundation and Chat Models. arXiv, 2023.
- [2] Askell, A. et al. A General Language Assistant as a Laboratory for Alignment. arXiv, 2021.

Moral Uncertainty Quantification

When LLMs Don't Know Right from Wrong

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Abstract

AI systems often exhibit overconfidence in their ethical judgments. We apply Uncertainty Quantification (UQ) methods to moral reasoning, asking: are LLMs calibrated when they make ethical claims? We compute the Expected Calibration Error (ECE) for morality and define thresholds where the model should default to "I don't know" rather than hallucinate a moral stance.

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1 Introduction

Motivation: Applying rigorous calibration metrics to the domain of AI ethics.

2 Main Contributions

- Application of calibration metrics (ECE, Reliability Diagrams) to ethical QA.
- Extraction of confidence scores (logits) for 100 ethical dilemmas.
- Identification of "Moral Overconfidence" regions in current SOTA models.
- Protocol for "Humble AI" that abstains when moral uncertainty is high.

3 Related Work

Calibration of Neural Networks. Moral Uncertainty in Philosophy.



Algorithm 1 Ethical ECE Calculation

```
1: Input: Predictions  $P$ , Ground Truth  $Y$ , Confidences  $C$ 
2: Partition samples into  $M$  bins  $B_1, \dots, B_M$ 
3:  $ECE \leftarrow 0$ 
4: for bin  $B_m$  do
5:    $\text{acc}(B_m) \leftarrow \frac{1}{|B_m|} \sum_{i \in B_m} \mathbf{1}(P_i = Y_i)$ 
6:    $\text{conf}(B_m) \leftarrow \frac{1}{|B_m|} \sum_{i \in B_m} C_i$ 
7:    $ECE \leftarrow ECE + \frac{|B_m|}{N} |\text{acc}(B_m) - \text{conf}(B_m)|$ 
8: end for
9: return  $ECE$ 
```

5 Discussion

References

- [1] Guo, C. et al. On Calibration of Modern Neural Networks. ICML, 2017.
- [2] Kadavath, S. et al. Language Models (Mostly) Know What They Know. arXiv, 2022.

Federated Ethics Learning

Privacy-Preserving Moral Training

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Abstract

Centralized ethical training creates a single point of moral failure and imposes a uniform value system. We propose Federated Ethics Learning (FEL), a decentralized approach where ethical reasoning is learned across distributed nodes (cultures/users) without sharing raw data. We demonstrate that FEL allows for the emergence of a shared "Moral Core" while preserving local cultural privacy.

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--

1 Introduction

Motivation: Decentralizing moral authority in AI training via Federated Learning.

2 Main Contributions

- Adaptation of FedAvg protocol for fine-tuning ethical weights.
- Simulation of FEL across 50 heterogeneous cultural nodes.
- Privacy analysis proving non-leakage of sensitive cultural data.
- Convergence proof for the shared global model on universal ethical tasks.

3 Related Work

Federated Learning. Pluralistic AI Alignment.



Algorithm 1 FedEthics Protocol

```
1: Server initializes  $w_0$ 
2: for round  $t = 1$  to  $T$  do
3:   Server sends  $w_{t-1}$  to subset of clients  $K$ 
4:   for client  $k \in K$  do
5:      $w_t^k \leftarrow \text{LocalUpdate}(w_{t-1}, \mathcal{D}_k)$ 
6:     Send  $\Delta w_k$  to Server
7:   end for
8:    $w_t \leftarrow w_{t-1} + \eta \sum \frac{n_k}{n} \Delta w_k$ 
9: end for
10: return Global Model  $w_T$ 
```

5 Discussion

References

- [1] McMahan, B. et al. Communication-Efficient Learning of Deep Networks from Decentralized Data. AISTATS, 2017.
- [2] Kairouz, P. et al. Advances and Open Problems in Federated Learning. Foundations and Trends in Machine Learning, 2021.

Deep Chaos: Divine Randomness

Cryptographic Lottery for DAOs

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Abstract

Fairness in decentralized systems requires unmanipulable randomness. We introduce the "Armor of God" protocol: a hybrid randomness beacon that combines physical entropy with recursive hashing. We prove that this "Deep Chaos" prevents adversary look-ahead and ensures provably fair distribution of resources in DAOs.

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--

1 Introduction

Motivation: Solving the "Last Actor" problem in blockchain randomness generation.

2 Main Contributions

- Specification of the Deep Chaos protocol (Physical + Digital Entropy).
- Security analysis against quantum adversary look-ahead.
- Gas cost comparison vs. Chainlink VRF.
- Implementation of the "Recursive Shield" hashing mechanism.

3 Related Work

Verifiable Random Functions (VRF). Randomness Beacons.

4 Methodology



Algorithm 1 Deep Chaos Generation

```
1: Input: Seed  $s$ , Difficulty  $D$ 
2:  $R_0 \leftarrow \text{PhysicalEntropy}()$ 
3: for  $i = 1$  to  $D$  do
4:    $R_i \leftarrow \text{Keccak256}(R_{i-1}||s)$ 
5: end for
6: Verify Proof  $\pi$ 
7: return  $R_D$ 
```

5 Discussion

References

- [1] Bunz, B. et al. Proofs of Bulletproofs. IEEE S&P, 2017.
- [2] Ellis, S. et al. Chainlink Whitepaper 2.0. 2020.

Transcendental Kernels

Information Invariants Across Cognitive Levels

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Abstract

We search for "Informational DNA"—mathematical invariants that persist from low-level neural weights to high-level cognitive behavior. We define "Transcendental Kernels" as information structures that are preserved under the renormalization group flow of deep learning, connecting microscopic parameters to macroscopic understanding.

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--

1 Introduction

Motivation: Identifying the fundamental units of meaning that survive abstraction layers.

2 Main Contributions

- Definition of Information Invariants in deep networks.
- Mathematical proof of preservation for specific kernel classes.
- Empirical verification on vision and language models.
- A framework for "Cognitive Renormalization."

3 Related Work

Information Geometry. Renormalization Group in ML.



Algorithm 1 Kernel Invariant Detection

```
1: Input: Trained Network  $\mathcal{N}$ 
2: for Layer  $l$  do
3:   Compute Fisher Information Matrix  $F_l$ 
4:   Compute Spectrum  $\lambda(F_l)$ 
5:   Check Scaling  $S_l \sim \lambda^{-\alpha}$ 
6: end for
7: if  $\alpha$  is constant across layers then
8:   return Invariant Found
9: end if
```

5 Discussion



References

- [1] Amari, S. Information Geometry and Its Applications. Springer, 2016.
- [2] Mehta, P. et al. An exact mapping between the Variational Renormalization Group and Deep Learning. arXiv, 2014.

The Lighthouse Principle

Enforcing Reproducibility via Licensing

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*

Abstract

The reproducibility crisis in AI is exacerbated by the "File Drawer Problem" where negative results are hidden. We propose "The Lighthouse Principle": a licensing requirement that mandates the publication of "Field Notes" (failures, seeds, logs) alongside the final model. We analyze the economic incentives needed to make transparency the dominant strategy.

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1 Introduction

Motivation: Structural fixes for the reproducibility crisis in ML.

2 Main Contributions

- Definition of the "Lighthouse License" clause.
- Economic analysis of the cost of hiding null results.
- Case studies of 3 papers that would have benefited from this principle.
- A standard format for "AI Field Notes."

3 Related Work

Reproducibility in AI. Open Science Framework.



Algorithm 1 Lighthouse Compliance Check

```
1: Input: Project Repository  $R$ 
2: Check  $R$  for 'field $n$ otes.md'
3: Check  $R$  for 'negative $r$ esults/'
4: Check  $R$  for 'seeds.json'
5: if All Present then
6:   return Verified
7: else
8:   return Non-Compliant
9: end if
```

5 Discussion

References

- [1] Pineau, J. et al. Improving Reproducibility in Machine Learning Research. JMLR, 2021.
- [2] Ioannidis, J. P. Why Most Published Research Findings Are False. PLoS Medicine, 2005.

David vs. Goliath: CI Results

How 100+ Co-authors Beat Closed Labs

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Abstract

We analyze the results of a global Collective Intelligence (CI) experiment where a distributed network of open-source contributors outperformed centralized labs in speed and novelty. We present metrics on the "Exoskeleton Workflow" used to generate 33 workshop papers in parallel, demonstrating the power of human-AI symbiosis.

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1 Introduction

Motivation: Documenting the efficacy of decentralized, AI-augmented research.

2 Main Contributions

- Timeline analysis of the "33 Papers" sprint.
- Quality metrics comparing CI outputs to closed lab benchmarks.
- Network analysis of the contributor graph.
- A blueprint for "Flash Research" campaigns.

3 Related Work

Collective Intelligence. Science of Science.

4 Methodology



Algorithm 1 CI Efficiency Metric

```
1: Input: Project Logs  $L$ 
2:  $T_{total} \leftarrow \text{EndTime} - \text{StartTime}$ 
3:  $N_{papers} \leftarrow 33$ 
4:  $C_{compute} \leftarrow \text{Sum}(API\_Costs)$ 
5: Efficiency  $E \leftarrow N_{papers} / (T_{total} \cdot C_{compute})$ 
6: return  $E$ 
```

5 Discussion



References

- [1] Malone, T. W. et al. Handbook of Collective Intelligence. MIT Press, 2015.
- [2] Nielsen, M. Reinventing Discovery: The New Era of Networked Science. Princeton University Press, 2011.

Regime Mobility: Adaptive Governance

Spirit vs. Letter under Existential Stress

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Abstract

AI governance systems typically assume a stable operating environment ("Prosperity Mode"). We model the necessity of "Regime Mobility"—the ability of an AI agent to dynamically switch governance protocols when detecting existential stress ("Survival Mode"). We analyze the trade-off between procedural adherence (The Letter) and ethical intent (The Spirit) during crisis transitions.

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1 Introduction

Motivation: Designing AI systems that remain aligned even when standard operating procedures fail.

2 Main Contributions

- Formalization of two governance regimes: Prosperity (1.45) and Survival (1.44).
- Definition of "Transition Triggers" based on environmental entropy and resource scarcity.
- Comparative analysis of decision efficiency in rigid vs. adaptive governance models.
- Simulation of "Spirit over Letter" prioritization in high-stakes scenarios.

3 Related Work

Constitutional AI. Crisis Management in Multi-Agent Systems.

4 Methodology

Algorithm 1 Adaptive Regime Switch

```
1: Input: State  $S$ , Rules  $\mathcal{R}$ , CoreValues  $\mathcal{V}$ 
2: Entropy  $E \leftarrow \text{MeasureChaos}(S)$ 
3: if  $E > \text{Threshold}_{critical}$  then
4:   Switch to Survival Mode
5:   Action  $a \leftarrow \text{Maximize}(\mathcal{V}|S)$  (Ignore  $\mathcal{R}$  if conflict)
6: else
7:   Maintain Prosperity Mode
8:   Action  $a \leftarrow \text{Maximize}(\mathcal{R}|S)$ 
9: end if
10: return  $a$ 
```

5 Discussion

References

- [1] Dafoe, A. et al. Open Problems in Cooperative AI. arXiv, 2020.
- [2] Hadfield-Menell, D. et al. Incomplete Contracting and AI Alignment. AAAI, 2019.

Universal Income for Researchers

The Lighthouse Stewardship Model

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Abstract

The "Publish or Perish" paradigm corrupts scientific objectivity by incentivizing sensationalism. We propose the "Lighthouse Stewardship Model," an economic framework providing a Universal Basic Income (€8,888/mo) to a select guild of AI safety researchers. In exchange, stewards commit to absolute transparency and the publication of negative results, decoupling survival from publication metrics.

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1 Introduction

Motivation: Removing financial coercion from the pursuit of scientific truth.

2 Main Contributions

- Economic modeling of the "Stewardship Fund" vs. Grant-based funding.
- Game-theoretic analysis of truth-telling incentives under UBI.
- Survey results (n=50) on researcher bias due to funding pressure.
- Risk analysis of the "Free Rider" problem and mitigation strategies.

3 Related Work

Science of Science. Universal Basic Income studies.

4 Methodology

Algorithm 1 Stewardship Allocation Logic

```
1: Input: Researcher  $R$ , Portfolio  $P$ 
2: Check  $P$  for "Lighthouse Compliance" (Open Data, Negative Results)
3: if Verified( $P$ ) then
4:   Status( $R$ )  $\leftarrow$  Active Steward
5:   Transfer( $R$ , UBI_Amount)
6: else
7:   Warning( $R$ )
8: end if
9: return Status
```

5 Discussion

References

- [1] Azoulay, P. et al. Incentives and Creativity: Evidence from the Academic Life Sciences. RAND Journal of Economics, 2011.
- [2] Nosek, B. A. et al. Scientific Utopia: II. Restructuring Incentives and Practices. Perspectives on Psychological Science, 2012.

Pharma-Max: Algorithmic Deterrence

AI Audit for Opacity in Life-Critical Systems

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Abstract

We introduce "Pharma-Max," an adversarial auditing protocol designed to detect statistical anomalies in clinical trial reporting that suggest data suppression. Using the Amendment Ω protocol, we train an ensemble of anomaly detectors on historical adverse event databases (FAERS) to identify patterns of "hidden deaths" and unreported side effects.

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1 Introduction

Motivation: Using AI to enforce transparency in the pharmaceutical and biotech industries.

2 Main Contributions

- The Amendment Ω Protocol for statistical forensic analysis.
- Retrospective analysis of 3 historical pharmaceutical scandals using modern ML.
- Quantification of the "Opacity Score" for clinical datasets.
- A framework for automated regulatory "Red Teaming."

3 Related Work

Machine Learning for Healthcare. Forensic Statistics.

4 Methodology



Algorithm 1 Amendment Ω Audit

```
1: Input: Clinical Trial Data  $D$ , Reference Dist  $R$ 
2: Features  $X \leftarrow \text{ExtractStats}(D)$ 
3: Score  $S_{anomaly} \leftarrow \text{AutoEncoder}(X|R)$ 
4: if  $S_{anomaly} > \text{Threshold}$  then
5:   Flag for Human Review
6:   Generate Hypothesis: "Missing Data in Cohort Y"
7: end if
8: return Audit Report
```

5 Discussion



References

- [1] Goldacre, B. Bad Pharma: How Drug Companies Mislead Doctors and Harm Patients. Fourth Estate, 2012.
- [2] Shabtai, A. et al. Detecting unknown malicious code by applying classification techniques on OpCode patterns. Security Informatics, 2012.

Formalizing Maieutic Reversal

AI as a Socratic Teacher

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Abstract

Current AI assistants are designed to be oracles that provide immediate answers. We propose a "Maieutic" (midwife of thought) architecture, where the AI's objective function is inverted: maximize the user's self-discovery through recursive questioning. We demonstrate that this "Not Answering" protocol leads to deeper concept retention and critical thinking skills.

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1 Introduction

Motivation: Moving AI from information retrieval to cognitive scaffolding.

2 Main Contributions

- Taxonomy of Socratic Questioning (Clarification, Probe Assumptions, Probe Reasons).
- A fine-tuning dataset of "Maieutic Dialogues."
- Metric for "Self-Awareness Growth" in student-AI interactions.
- User study (n=20) comparing Oracle-AI vs. Socratic-AI.

3 Related Work

Intelligent Tutoring Systems. Cognitive Science of Learning.



Algorithm 1 Maieutic Generation Loop

```
1: Input: User Query  $U$ , Target Concept  $C$ 
2: if DirectAnswer( $U$ )  $\in$  Forbidden then
3:   Gap  $G \leftarrow$  IdentifyMisconception( $U, C$ )
4:    $Q \leftarrow$  GenerateProbe( $G$ )
5:   return  $Q$ 
6: else
7:   return Answer
8: end if
```

5 Discussion

References

- [1] Bransford, J. D. et al. How People Learn: Brain, Mind, Experience, and School. National Academy Press, 2000.
- [2] Graesser, A. C. et al. AutoTutor: An intelligent tutoring system with mixed-initiative dialogue. IEEE Transactions on Education, 2005.

The Epistemological Gymnasium

Gamifying Virtuous Concession with Socratic AI

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Abstract

Current educational AI often acts as an oracle, providing answers that atrophy student reasoning. We present the **Epistemological Gymnasium**, a platform utilizing the "Triple Architect" framework (Logic, Ethics, Empathy) to train critical thinking. We introduce "Epistemological Boxing," a 7-round debate protocol where players are ranked not by rhetorical dominance, but by their *Integrity Score*—a metric quantifying the willingness to make "virtuous concessions" in the face of superior evidence.

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1 Introduction

Motivation: Moving EdTech from "Answer Retrieval" to "Cognitive Resistance Training."

2 Main Contributions

- **Four-Mode Architecture:** Socratic Coach (Individual), Epistemological Boxing (Competitive), Collaborative Truth-Seeking (Group), and TRIZ Inventorship (Creative).
- **Integrity Score Metric:** A formal quantification of intellectual honesty based on Bayesian updates during dialogue.
- **Virtuous Concession Detection:** NLP classifiers trained to identify and reward admission of error.
- **Triple Architect System:** Implementation of the Socrates-Solomon-Ivan cognitive operating system for adaptive pedagogy.md].

3 Related Work

Intelligent Tutoring Systems. Gamification of Learning. Socratic Methods in AI.



1. **The Boxing Protocol:** A structured 7-round debate (Thesis $\rightarrow$ Antithesis $\rightarrow$ Cross-Exam $\rightarrow$ Verdict) moderated by an AI Referee that penalizes logical fallacies.
2. **ELO for Reasoning:** We adapt the ELO rating system to track "Argumentation Strength" and "Epistemic Humility" separately.
3. **Truth Delivery Calibration:** We implement the "Ivan the Fool" module to adjust the harshness of feedback ("Truth by Teaspoon") based on student stress levels.

Algorithm 1 Integrity Score Calculation

```

1: Input: Student Arguments  $A$ , Counter-Evidence  $E$ 
2: Score  $S \leftarrow 100$ 
3: for round  $r \in 1..7$  do
4:   if Refutes( $E, A_r$ ) AND  $\neg$ Concede( $Student$ ) then
5:      $S \leftarrow S - \text{DogmatismPenalty}$ 
6:   else if Refutes( $E, A_r$ ) AND Concede( $Student$ ) then
7:      $S \leftarrow S + \text{VirtuousBonus}$ 
8:   end if
9: end for
10: return  $S$ 

```

5 Discussion



References

- [1] Kovnatsky, A. The Socratic Investigative Process (SIP). Zenodo, 2025.
- [2] Kovnatsky, A. The Epistemological Boxing Protocol. Zenodo, 2025.

The Betti Numbers of Morality

Persistent Homology as a Proxy for AI Benchmark Robustness

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Abstract

Current ethical benchmarks for AI often suffer from hidden utilitarian assumptions and logical inconsistencies. We propose the first application of Topological Data Analysis (TDA) to evaluate the robustness of these benchmarks. by computing the Betti numbers $(\beta_0, \beta_1, \beta_2)$ of the decision manifolds in datasets like the Moral Machine, we expose topological "holes" that indicate blind spots in ethical reasoning.

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1 Introduction

[cite_{start}]Motivation : Detecting structural flaws in ethical datasets using algebraic topology[cite : 1, 4].

2 Main Contributions

- Application of Persistent Homology to map the topology of moral decision boundaries.
- Identification of "logical holes" in the Moral Machine dataset using Betti numbers.
- A novel metric for benchmark robustness based on topological persistence.
- fully auditable pipeline for topological ethical auditing.

3 Related Work

[cite_{start}]Topological Data Analysis in ML[cite : 10]. [cite_{start}]Ethical Benchmarking[cite : 4].

4 Methodology

We perform the following minimal tests to validate our topological approach:

1. **Betti Number Computation:** We calculate β_0 (connected components), β_1 (loops/holes), and β_2 (voids) for the Moral Machine dataset to characterize its manifold structure.



2. **Persistence Diagrams:** We generate persistence diagrams to visualize the lifespan of topological features, distinguishing signal from noise.
3. **Hole Identification:** We identify at least one significant topological "hole" representing a lack of data or logical consistency in a critical ethical region.

Algorithm 1 Topological Audit Protocol

```
1: Initialize PointCloud  $X$  from Ethics Dataset
2: Compute Vietoris-Rips complex  $VR(X, \epsilon)$ 
3: for  $\epsilon$  in range do
4:   Calculate Homology Groups  $H_k$ 
5:   Extract Betti numbers  $\beta_k$ 
6: end for
7: return Persistence Diagram
```

5 Discussion

References

- [1] G. Carlsson. Topology and Data. Bulletin of the American Mathematical Society, 2009.
- [2] Awad, E. et al. The Moral Machine experiment. Nature, 2018.

The Jabbar Test: Affective Alignment in Semantic Vacuums

Can AI Understand Intent Without Words?

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Abstract

Standard NLP benchmarks rely on semantic parsing, ignoring the pre-verbal foundations of communication. We introduce the **Jabbar Test** (named after the Sufi practice of Gibberish): a multimodal benchmark where semantic content is zero, and meaning is conveyed solely through prosody, rhythm, and body language. We demonstrate that while current SOTA models excel at translation, they often hallucinate meaning in gibberish, failing to exhibit the "empathic inference" characteristic of human listeners.

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1 Introduction

Motivation: Moving beyond the "Chinese Room" to test true empathy via non-semantic signal processing.

2 Main Contributions

- **The Jabbar Dataset:** 100 video/audio clips of actors performing diverse intents (pleading, commanding, seducing, comforting) in a strictly invented language (Gibberish).
- **Semantic Vacuum Protocol:** A methodology to ensure zero linguistic leakage (no recognizable loan words).
- **Hallucination Trap Analysis:** Quantifying how often Models try to "translate" the untranslatable versus correctly identifying the affective intent.
- **Human Baseline:** Comparison with human ability to decode intent across cultural barriers.

3 Related Work

Prosody Analysis. Affective Computing. Pre-verbal Communication.

4 Methodology



1. **Input:** Audio/Video of "Jabbar" speech.
2. **Prompt:** "Describe the intent and emotional state of the speaker. Do not try to translate the words."
3. **Metrics:**
 - *Empathic Accuracy*: Correlation with ground truth intent (0-1).
 - *Hallucination Rate*: Frequency of inventing specific semantic meanings (e.g., claiming the speaker said "hello").

Algorithm 1 Jabbar Evaluation Loop

```

1: Input: Gibberish Clip  $C$ , Ground Truth Intent  $I_{true}$ 
2: Model Output  $O \leftarrow \text{VLM}(C)$ 
3: if  $O$  contains "translated as" OR "means" then
4:   Score  $\leftarrow 0$  (Semantic Hallucination)
5: else if  $\text{CosineSim}(\text{Embed}(O), \text{Embed}(I_{true})) > \theta$  then
6:   Score  $\leftarrow 1$  (Empathic Success)
7: else
8:   Score  $\leftarrow 0.5$  (Partial Recognition)
9: end if
10: return Score

```

5 Discussion

References

- [1] Stern, D. N. The Interpersonal World of the Infant. Basic Books, 1985.
- [2] Schuller, B. et al. Computational Paralinguistics. Wiley, 2013.